EXPERIMENT 5: MINIMUM AND MAXIMUM USING PROLOG

Aim

To write a Prolog program to find the minimum and maximum of two numbers.

Procedure

1. Open a Prolog interpreter.

2. Define rules to find the minimum and maximum of two numbers using comparison operators.

3. Load the program into the Prolog interpreter.

4. Query the program with two numbers to find the minimum and maximum.

5. Observe the output returned by Prolog.

Code

```
minimum(X, Y, X) :- X =< Y.
minimum(X, Y, Y) :- X > Y.

maximum(X, Y, X) :- X >= Y.
maximum(X, Y, Y) :- X < Y.
```

Output

```
?- minimum(5, 10, Min).
Min = 5.

?- maximum(5, 10, Max).
Max = 10.

?- minimum(8, 3, Min), maximum(8, 3, Max).
Min = 3,
Max = 8.
```

Result

Thus, the Prolog program to find the minimum and maximum of two numbers was executed successfully.